CH-1101: Quantum Chemistry and Spectroscopy

Tutorial Sheet 8

Module: Molecular spectroscopy

1) What changes in the characteristics (*position*, *intensity* or *width*) of a spectroscopic signal are expected when the _____?
   (a) excited state to ground state energy gap increases
   (b) lifetime of the excited state decreases
   (c) probability of transition between the two state decreases

2) Two spectroscopic transitions from an initial (ground) state to a final (excited) state exhibit line widths of (a) 100 MHz and (b) 10 cm^{-1}, respectively. Assuming no Doppler broadening is present, estimate the lifetime (δt) of these two excited states.

3) The molar absorption coefficient of an organic dye molecule dissolved in hexane is known to be 227 L mol^{-1} cm^{-1} at 500 nm. Calculate the percentage reduction in intensity of light when light of wavelength 500 nm passes through 2.00 mm long sample cell containing a solution of this organic dye molecule of concentration 2.52 x 10^{-3} mol L^{-1}.

4) When light of 600 nm wavelength is passed through a 6.50 mm long sample cell containing an aqueous solution of an organic compound (molecular weight: 200 g mol^{-1}), 48.3% of the light is absorbed. Determine the amount (in mg) of this organic compound present in 1 L of this solution. The molar absorption coefficient (ε) of this organic molecule at 600 nm is 423 L mol^{-1} cm^{-1}.

5) A molecule consists of two quantized energy states (a ground state and an excited state), separated by an energy gap of ΔE. In a large collection of this molecule, just a very small fraction of 0.001% of molecules occupy the excited state even when the temperature was raised to 500 K. Determine the wavelength of light (in nm) this molecule will absorb to cause a transition from the ground state to the excited state.

6) What kind of spectroscopic transitions are probed by each of the following types of electromagnetic radiations?
   (a) Ultraviolet (UV) – visible (b) Infrared (IR) (c) Microwaves (d) Radio waves

Fundamental constants:	*Mass of electron (m_e) = 9.109 x 10^{-31} kg*
Speed of light (c) = 2.997 x 10^{8} m s^{-1}	*Mass of proton (m_p) = 1.672 x 10^{-27} kg*
Elementary charge (e)= 1.602 x 10^{-19} C	*Plank's constant (h) = 6.626 x 10^{-34} J s*
Avogadro's number (N_A) = 6.023 x 10^{23} mol^{-1}	***Conversion factor:***
Boltzmann constant (k_B) = 1.381 x 10^{-23} J K^{-1}	*1 eV = 1.602 x 10^{-19} J*